Session 1 Materials-Engineering:

Tuesday, 8th of September 2026, 12.45-2.15 (1.30 h), National Research Center.

Prof. Ahmed A. Abdel Moneim is an internationally recognized research leader, materials scientist, and higher education executive with more than 25 years of experience leading institutional transformation, building globally competitive research ecosystems, and advancing innovation-driven research and technology commercialization. His research interests encompass advanced materials, graphene and other two-dimensional materials, heterogeneous catalysis, printed electronics for IoT applications, flexible sensors, clean energy technologies, and sustainable manufacturing. Throughout his career, he has established academic programs, multidisciplinary research centers, advanced laboratories, and strategic international partnerships that have strengthened institutional research capacity, fostered interdisciplinary collaboration, and translated scientific discoveries into technologies and solutions with tangible industrial and societal impact.

As Founding Dean of the Faculty of Basic and Applied Sciences at the Egypt-Japan University of Science and Technology (E-JUST), Prof. Abdel Moneim also founded the Department of Materials Science and Engineering, the Graphene Center of Excellence for Energy and Electronic Applications, and the Industrial Catalysis Laboratory. Under his leadership, E-JUST expanded its research capabilities through the establishment of advanced research infrastructure, the development of high-impact research programs, and strategic collaborations with leading universities and research institutions, including Virginia Tech (USA); Tohoku University, Tokyo Institute of Technology, and the National Institute for Materials Science (NIMS) (Japan); TU Dresden and the University of Kassel (Germany); as well as leading research institutions in China and South Korea. He also established long-term partnerships with major industrial organizations, including EZZ Steel, Pharco Pharmaceuticals, Abu Qir Fertilizers and Chemical Industries, ETHYDCO (The Egyptian Ethylene and

Derivatives Company), and MIDO Coatings, accelerating the translation of research into advanced materials, catalysis, printed electronics for IoT applications, clean energy technologies, sustainable manufacturing, and commercially relevant innovations.

Prof. Abdel Moneim has authored more than 160 peer-reviewed publications, holds four granted international patents, has filed more than ten patent applications, secured over EGP 65 million in competitively awarded national and international research funding, and supervised more than 67 M.Sc. and Ph.D. graduates. His contributions to research and innovation have been recognized through the State Excellence Award in Advanced Technological Sciences (2024), inclusion among Stanford University's World's Top 2% Scientists (2020–2025), and the UNDP Invention for Humanity Award (2025).

Prof. Abdel Moneim believes that research excellence is measured not only by publications and patents, but by its ability to generate innovation, develop the next generation of scientific leaders, strengthen industrial competitiveness, inform public policy, and create lasting value for society.

Turning CO_2 into Opportunity: The New Carbon Economy

The transition from carbon-intensive processes to a circular carbon economy requires technologies that can transform CO_2 from an emissions burden into a valuable carbon resource. Catalytic hydrogenation of CO_2 offers a promising pathway for converting captured CO_2 into hydrocarbons and chemical feedstocks while integrating renewable hydrogen into carbon recycling processes. In this study, alkaline-promoted $ZnFe_2O_4$ catalysts were developed to investigate how surface basicity, reducibility, and process configuration can be engineered to improve CO_2 conversion and hydrocarbon production. A 1.5%Na–$ZnFe_2O_4$ catalyst exhibited strong basic character, with a total surface basicity of 0.261 mmol g^{-1}, and demonstrated effective CO_2 activation under high-pressure reaction conditions. At 330–340 °C, 25 bar, a gas hourly space velocity of 2245 mL g^{-1} h^{-1}, and an H_2/CO_2 ratio of 3:1, the catalyst achieved approximately 41% CO_2 conversion, with hydrocarbons dominated by olefins (~63%), highlighting its potential for producing higher-value carbon products rather than predominantly methane.

Dual alkaline promotion through the incorporation of potassium further increased CO_2 conversion to approximately 49%; however, the enhanced hydrogenation activity shifted product distribution toward lighter hydrocarbons and increased methane and C_1–C_4 selectivity. Beyond catalyst design, process intensification through tail-gas recycling provided a substantial improvement in carbon utilization. Recycling the unreacted reactor effluent through a secondary catalytic stage increased overall CO_2 conversion from approximately 41% to 75% and enhanced the hydrocarbon yield from 1.04×10^{-2} to 1.91×10^{-2} mol g^{-1} h^{-1}, while reducing CO and methane formation and maintaining high olefin production.

These findings demonstrate that the emerging carbon economy can be advanced through the integration of catalyst engineering and process-level carbon management. Alkaline-promoted $ZnFe_2O_4$ enables the selective transformation of CO_2 into value-added hydrocarbons, while tail-gas recycling substantially improves carbon conversion and resource efficiency. The combined strategy provides a promising foundation for moving beyond CO_2 mitigation toward CO_2 valorization, carbon circularity, and the production of marketable carbon-based products, positioning captured CO_2 as a feedstock for a new generation of sustainable chemical and energy technologies.

Prof. Dr. Nour F. Attia is Full professor at gas analysis and fire safety lab, chemistry division, National Institute of standards (NIS), Prof. Attia's research contributions in the field of nanochemistry, energy gasses (H2 & CH4) storage, gas separation and greenhouse gases capture, smart textile, flame retardants and water treatments. This is in addition to sensors fabrication and thermal energy storage media. Prof. Attia published more than 116 international papers in high impact factor journals and presented in more than 60 in international conferences. Additionally, Dr. Nour published 5 book chapters, 6 international patents (2 USA and 4 Korean) and six Egyptian patents were recently filled. His H-index is 42. Also, he has been awarded several national and international prizes such as State Prize, National Institute of Standards Prize, Medal of Excellence from the first class from President of the Republic 2020 , Obada-Prize and Climate Change Alleviation Award 2022 from Mansoura University. He has been elected as a member of the Egyptian Young Academy of Sciences for 2021-2024. Additionally, he has selected among the highest distinguished top 2% worldwide scientists in the following connective years 2021,2022,2023,2024 and 2025. He has elected as member of National committee of new and advanced materials 2022-2025 and elected for another term 2025-2028. He is PI of several research projects funded from ASRT and STDF. Interestingly, he organized and chaired various international conferences and workshops in the filed of hydrogen energy, greenhouse gases and nanotechnology. Additionally, Prof. Attia academically supervised more than 20 postgraduates' students and teaching in different Egyptian universities. Recently, he has been selected as member of Electricity and Energy Research Council 2025-2028, ASRT, Egypt. Also, he got the Prof. Dr. President of The Research Staff Club Award in Basic Science and Technological Applications, NRC, Egypt (Oct. 2025).Also, he is editor-in-chief of Journal of Climate Change Action and editorial board members for many international Journals.

Sustainable and Scalable Nanomaterials for Energy and Environmental Applications

Energy and environmental pollution crises were anticipated as result of rapid consumption of limited resources fossil fuels over the world in various purposes. Hence, the demand for clean and renewable alternative source of energy is necessity. Hydrogen (H2) is considered as alternative green, emission less, renewable and high-density energy source for vehicles. Nevertheless, the promising merits of hydrogen as future fuel for vehicles, its commercialization is still tied due to its gases state which technically, restricted its safe and efficient storage. Hence, the need for efficient and safe solid-state hydrogen storage media is the main challenge for commercialization of hydrogen energy. Thus, the scalable production of efficient nanomaterials is crucial for overcoming this energy crisis and other environmental challenges such as climate change crisis and other water crisis. Therefore, this lecture will cover the different routes for scalable synthesis of nanomaterials and their utilization for hydrogen storage applications, CO2 capture and utilization and other environmental applications.

Marwa Shalaby is a Professor of Chemical Engineering at the National Research Centre in Cairo, Egypt, with a Ph.D. from Cairo University. She has extensive teaching experience and has conducted research in areas like water desalination, nanotechnology, and wastewater treatment. Marwa has published over 90 research articles and holds several patents related to membrane technology. Her professional experience includes roles as a researcher and project leader in various international collaborations. She has led significant research projects funded by various national and international bodies, emphasizing sustainable development in water resources. She has received multiple awards for scientific excellence and actively supervises Ph.D. and M.Sc. students. Prof. Shalaby is also an invited/keynote speaker at numerous national and international conferences.

Next-Generation Composite Membranes for Water and Wastewater Treatment

Rapid industrialization, urban expansion, and the intensification of agricultural and pharmaceutical activities have accelerated the release of emerging pollutants into ecosystems, posing significant threats to environmental and public health. Among these contaminants, nicotine—a toxic and highly soluble alkaloid originating from tobacco production, cigarette waste, and pharmaceutical residues—has increasingly been detected in surface water, raising urgent concerns regarding aquatic pollution and long-term ecological impacts. This study investigates the enhancements in the removal of emerging pollutants through the utilization of advanced membrane technologies. The incorporation of chitosan into polyethersulfone (PES) membranes demonstrates significant benefits, including improved dye removal efficiency, filtration performance, and fouling resistance. Chitosan-modified membranes maintain a stable dye removal efficiency over time, while increasing permeability to support higher water flow rates and reduced pressure requirements, crucial for large-scale applications like desalination. In addition, novel mixed-matrix nanofiltration membranes have been developed to effectively separate and recover nicotine from aqueous media. These membranes, integrating hydrophilic functional composites into a polyvinyl chloride (PVC) matrix, leverage operational parameters such as nicotine concentration, temperature, and pH to optimize removal efficiency. Membranes modified with tannic acid (TA) and polyvinylpyrrolidone (PVP) achieved up to 75% nicotine removal at an initial feed concentration of 80 mg/L, with a remarkable flux recovery rate of 97.65% following humic acid fouling. The study also analyses three membrane types: M1, M2, and M3. M1 shows strong initial antifouling performance but declines quickly, making it suitable for short-term applications. In contrast, M2 offers a balanced decrease in antifouling effectiveness, ideal for moderate to long-term use. M3 exhibits high initial resistance but lacks durability, limiting its prolonged effectiveness. Structural and physicochemical characterization confirms the enhancement in functional performance, with mechanical strength increasing from 40.7 to 58.4 MPa and contact angle decreasing to 54.1°. Overall, the advancements in both chitosan-modified PES membranes and functional composite membranes highlight their potential as sustainable and efficient materials for addressing emerging pollutants like nicotine and dyes in surface waters. Future research should focus on optimizing these materials and assessing their long-term stability under various environmental conditions.

Prof. Dr. Ahlam M. Fathi is currently working as a professor of applied physical chemistry at physical chemistry department, National Research Centre (NRC), Egypt. She received his PhD (2004) from Chemistry department, faculty of science, Cairo University. She has over 25 years of research experience, teaching, training and student supervision in the field of electrochemistry, corrosion protection, electrocatalysis, photochemistry, supercapacitors and Energy production. In 2009, she worked as associate prof. in chemistry dept., faculty of science, King Faisal university, KSA. She is author/ co-author of over 54 scientific articles in peer-reviewed journals. She is a potential reviewer of more than 40 scientific articles and student projects. She is a supervisor for 4 MSc and pHD students.

Progress in using metal oxide-based electrode materials as safe and sustainable electrode for electrochemical supercapacitors

In the modern world, energy drives all the daily services starts from production to transportation, though generation of renewable energy sources grows. Although renewable sources can generate substantial energy, large-scale storage remains a major challenge. Economic pressures have accelerated the development of effective electrochemical energy storage devices (EESDs), such as rechargeable batteries, fuel cells, and supercapacitors (SCs). Among these, researchers have recently highlighted SCs as a highly viable alternative due to their unique performance characteristics, which include high power density (PD), scalable energy density (ED), rapid charge-discharge capabilities, and excellent cyclic stability. Modifying metal oxides (MOs) with using nanotechnology is reviewed to obtained high performance electrode materials for SCs. Therefore, several feasible synthesis processes for preparing metal oxide (MO) nanoparticles and their composites, and discussing how MO morphology affects electrochemical performance are highlighted. The novelty of these MO electrode materials stems from their rich redox activity, excellent conductivity, and strong chemical stability, as well as the synergistic effects present in their hybrid forms, making them outstanding candidates for SC applications.

Dr. Tamer Sami Mansour is an Associate Research Professor in the Department of Refractories, Ceramics, and Building Materials at Egypt's National Research Centre (NRC). He earned his Ph.D. in Chemistry from Ain Shams University, focusing his pioneering work on utilizing local Egyptian diatomite for heat-insulating materials. With more than two decades of academic and industrial expertise, Dr. Mansour is a leading researcher in advanced composites, porous ceramics, and sustainable waste recycling. His extensive contributions to the scientific community include a robust portfolio of international publications dedicated to ecological preservation and advanced materials innovation.

Diatomite-based electromagnetic wave absorption and shielding materials: Current research and future possibilities

Diatomite is a soft, siliceous sedimentary rock formed naturally from the fossilized remnants of diatoms, which are unicellular algae with a high natural amorphous silica content (80 to 90% silica). Diatomite pores are naturally occurring, fine, and plentiful apertures that contribute to the siliceous rock's high porosity (60-80%) and large specific surface area (30-160 m2/g). These pores, which range in size from nanometers to microns (1.7 to 800 nm), are what give diatomite its outstanding absorption, filtration, and insulating capabilities.

In recent decades, diatomite has found new applications in sophisticated biomaterials, environmental science, and biomedicine, in addition to its conventional functions. Its distinct characteristics make it a candidate as a base biomaterial for electromagnetic wave-absorbing composite materials. This critical review article provides an overview of progress in recognizing the potential of diatomite composite materials, including 2D and 3D materials, and the different techniques for fabricating diatomite-electromagnetic wave-absorbing material composites (electromagnetic interference shielding materials), their applications, and prospects.

Dr. Heba Elsayed Ali Hassan is an Associate Professor of Physical Chemistry at the National Research Center (NRC) in Egypt, where she has established a robust career in advanced materials and environmental sustainability. She earned her PhD in Chemistry from Ain Shams University in 2016, specializing in the synthesis of modified TiO2 arrays for solar energy conversionâ€”a research endeavor that was honored with the Best PhD Thesis Award in 2017. Her research systematically explores the kinetics and isotherms of pollutant adsorption, the photocatalytic degradation of organic dyes under solar illumination, and the fabrication of functional polymer nanocomposite membranes. Dr. Hassanâ€™s research expertise lies at the intersection of nanotechnology, photocatalysis, and renewable energy. Her work focuses on the strategic design of 0D, 1D, and 2D nanocomposites, such as CdO/Yâ‚‚Oâ‚ƒ/graphene oxide and g-Câ‚ƒNâ‚„/CdS/ZrOâ‚‚ to facilitate solar-driven hydrogen production and the degradation of hazardous industrial pollutants. A prolific contributor to the scientific community, Dr. Hassan has authored numerous peer-reviewed publications in prestigious international journals and serves as a regular reviewer for prominent publishers, including Elsevier, Springer, and Wiley. Her professional leadership extends to organizing international workshops and mentoring undergraduate students. She is also an active member of several esteemed organizations, including the Society for Women in Science in Developing Countries (OWSD) and the Egyptian Society of Advanced Materials and Nanotechnology.

Advanced Ternary Nanocomposite Heterojunctions for Solar-Driven Hydrogen Evolution and Environmental Remediation

Addressing the intersecting challenges of global energy scarcity and widespread water pollution requires the engineering of highly efficient, visible-light-driven photocatalysts capable of facilitating sustainable fuel production and environmental remediation. While traditional single-semiconductor photocatalysts are vital, they are fundamentally constrained by restricted light absorption and the rapid recombination of photogenerated charge carriers. This lecture explores the design of high-performance and reusable ternary photocatalysts; including 0D/1D/2D CdO/Y2O3/graphene oxide, 2D/0D g-C3N4/CdS/ZrO2, and Ag3PO4/NiO/TiO2 mesoporous p-n heterojunctions. These advanced systems utilized transformative potential for green fuel production and wastewater treatment. Experimental results demonstrate exceptional synergy, achieving near-complete degradation of hazardous industrial dyes (Crystal Violet, Methylene Blue, and Malachite Green) and impressive solar hydrogen evolution rates. Specifically, CdO/Y5/GO40 exhibited a dye degradation rate constant of ~2.4, 22, 1.5, and 4.9 times higher than CdO, Y2O3, CdO/Y5, and P25, respectively. g-C3N4/CdS/ZrOâ‚‚ system exhibits a rate constant 13.5 times greater than pure ZrOâ‚‚ and 1.8 times greater than its binary counterpart. Agâ‚ƒPOâ‚„/NiO/TiOâ‚‚ triheterojunction achieved a degradation rate constant 70 times higher than pristine TiOâ‚‚ and 20 times higher than the binary NiO/TiOâ‚‚ mixed oxide. Quantitative analysis reveals that the CdO/Y2O3/GO system achieves a remarkably narrow bandgap of 1.45 eV, a high surface area of 160.57 mÂ²/g and significantly quenched photoluminescence intensity, confirming inhibited recombination, consequently facilitating a hydrogen evolution rate of 2410 Î¼mol/g/h. g-C3N4/CdS/ZrO2 heterojunction, utilizing a dual S-scheme charge transfer pathway, yields an Hâ‚‚ production rate of 3928.32 Î¼mol/g/h. Notably, the Ag3PO4/NiO/TiO2 triheterojunction demonstrated the highest Hâ‚‚ evolution rate of 4642.56 Î¼mol/g/h.

Session 2 Materials-Engineering:

Tuesday, 8th of September 2026, 3.00-5.00 (2 hours), National Research Cener.

Giovanna Avellis is leading researcher in ICT at InnovaPuglia SpA, Italy, and has been involved as project manager in a number of European projects in Software Engineering and Mobile Telecommunication. She served the EU Commission as independent expert evaluator. Her current research interest is in Gender Equality and Artificial Intelligence. Currently scientific reviewer of several International Journal on mobile technologies.

She is Delegate of Women20 Italy since 2019, an engagement group of G20 on Gender Equality policies. Director of iCORSA (International Cornsortium of Research Staff Associations) www.icorsa.org , she founded and is chairing the Italian branch of ICoRSA ItalianRSA (Italian Research Staff Association). Past President of ITWIIN (Italian Women Innovators and Inventors Network) www.itwiin.org and currently vice- President of ITWIIN APS (Associazione di Promozione Sociale). Founder of two Working Groups on Gender Equality in Marie Curie Fellows Association and Marie Curie Alumni Association Advisory Board member of the Italian CNR Observatory GETA (Gender and Talent).Member of Italian women associations Donne&Scienza and CREIS (European Center for Sustainable Innovation).

Entrepreneurship and Science innovation by tackling the Women Digital Divide Gap

Entrepreneurship and Financial Inclusion of women are key issues to be tackled: improving gender balance in entrepreneurship could have a 6% return on investment up to $5 trillion. This will increase local job creation, tax revenues, boost innovation, drive macroeconomic gains and contribute to community well-being by reinvesting in education, health, and family welfare. Another important issue is to accelerate entrepreneurship ecosystems with inclusive legal, policy, and regulatory frameworks to provide enhanced opportunities for women entrepreneurs from start-up to scaling through growth. To address this issue, we need to secure equal rights to asset ownership and finance and develop tailored programs for women in business training centres to include education and mentoring in finance, entrepreneurship, and emerging sectors (artificial intelligence, blue and green economies, care economy, and space). Science innovation can be facilitated investing in Education, STEM and the Digital Divide: implementation and financing gaps persist. These gaps are underpinned by harmful social stereotypes and norms that impact women and girls at all levels of education, especially in the STEM fields, as well as in an increasingly AI-driven digital economy. Finally, it is advising to mandate targeted national scholarships and funding schemes for women and girls -especially those from marginalised communities - pursuing high-growth STEM fields such as AI, electronic engineering and informatics, and digital innovation spanning the course of education to employment.

"Degree in Computer Science, about 100 papers in peer reviewed conferences and journals, 30 years in industrial research and 4 years in academic research, and 18 months at Imperial College, London, UK, Distributed Software Department, funded by a Marie Curie Individual Fellowship.

Prof.Dr. Shimaa El-Hadad is a Professor and Vice Dean of the Manufacturing Technology Institute at Central Metallurgical Research and Development Institute (CMRDI), Egypt. Dr. Shimaa has 26 years of experience in Manufacturing generally and metal casting specifically. She has obtained her Ph.D. in materials engineering from Nagoya Institute of Technology, Japan, and her M.Sc. from the University of Quebec, Canada. She published more than 85 articles and managed several projects that targeted the localization of the spare parts industry. She used to teach training courses to engineers from the industry on casting and heat treatment of metallic alloys.

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Titanium Alloys: Recycling Technologies and Challenges

With the rising global demand for clean and efficient energy solutions, there is a growing need for materials capable of performing under extreme conditions while improving both performance and durability. Titanium has become a key material in the energy sector due to its excellent corrosion resistance, high strength-to-weight ratio, and ability to withstand severe temperatures and pressures. Its use spans nuclear reactors, offshore oil and gas platforms, and renewable energy technologies, including solar, wind, and hydrogen systems, highlighting its critical role in advancing energy infrastructure and shaping the future of power generation. This review presents a market analysis of global titanium production, its various applications, and the state of titanium scrap worldwide. It also provides a brief overview of the technologies and challenges involved in recycling titanium, particularly in the context of renewable energy. Lastly, promising future opportunities in titanium recycling are introduced.

Prof. Dr. May M. Eid is a Professor of Biophysics in the Spectroscopy Department of the Physics Division at Egypt's National Research Centre (NRC) in Cairo, where her research sits at the intersection of vibrational spectroscopy, green nanotechnology, and cancer nanomedicine.

She trained as a biophysicist at Cairo University before earning a master's degree in physics at Ain Shams University. She went on to complete a Ph.D. in Radiation Oncology and Biology through a joint program between the University of Oxford's Gray Institute and Ain Shams University, supported by a Ministry of Higher Education fellowship. Her postdoctoral work took her to the Cancer Centre at the University of Medicine and Dentistry of New Jersey on an STDF fellowship, and later to the SISSI beamline at the Elettra Synchrotron in Trieste, Italy, where an ICTP TRIL fellowship allowed her to train in synchrotron-based SR-FTIR microscopy under Lisa Vaccari.

Dr. Eid joined the NRC in 2005 and has advanced through its research ranks to Professor, a position she has held since 2023. Her work centers on using FTIR and SR-FTIR spectroscopy as a diagnostic lens on biological and nanomaterial systems — tracking how green-synthesized, chitosan-functionalized nanoparticles (silver, zinc oxide, and superparamagnetic iron oxide core-shell structures) interact with cells and tissue. This spectroscopic approach underpins her broader research program in nano-enabled cancer therapeutics, including sorafenib- and microRNA-loaded nanocarriers developed and evaluated for hepatocellular and colorectal cancer, as well as immunogenicity studies of nanocomposite vaccine-adjuvant platforms. A parallel line of her research applies the same green-synthesis and characterization methods to agricultural and environmental problems, including bio-fungicides and nanomaterial risk assessment.

Dr. Eid has authored more than 30 peer-reviewed publications and book chapters, holds two Egyptian patents for nanocomposite synthesis methods, and has served as principal investigator on

STDF- and NRC-funded grants, as well as a joint CNR Italy–NRC Egypt project. She has supervised multiple doctoral and master's students at the NRC and co-organized several international spectroscopy conferences hosted by the Centre. She is fluent in Arabic, English, French, and Italian.

Green-Synthesized Magnetic-Plasmonic Core-Shell Nanoparticles for Targeted Drug Delivery

The use of Fourier Transform Infrared (FTIR) microscopy as a label-free, non-destructive method for identifying molecular and chemical alterations in biological samples treated with superparamagnetic iron oxide nanoparticles (SPION) nanocomposites will be discussed in this talk. We examined cellular and tissue samples after treatment to find notable biomolecular changes, such as changes in lipid, protein, and nucleic acid content, by utilizing the sophisticated spatial resolution and spectral sensitivity of FTIR microscopy. Our approach combines multivariate data analysis with hyperspectral imaging to provide a comprehensive chemometric evaluation of nanoparticle-cell interactions and their biological effects. The findings support SPION nanocomposites' potential use in determining their effectiveness in certain therapeutic and diagnostic applications by showing that they cause quantifiable biochemical changes. These results highlight the usefulness of FTIR microscopy in preclinical research for assessing the biocompatibility of new nanomaterials and in nanotoxicological evaluations.

Dr. Nihal Ahmed El-Mehalawy is an associate professor at Refractories, Ceramics and Building Materials department, Advanced Materials Technology and Mineral Resources Research Institute, National Research Centre, Egypt. She gained her B. Sc., M. Sc. and Ph. D degrees from Faculty of

Sciences, Ain Shams University. Her fields of interests include Ceramic Science and Technology, Nanotechnology, Biomaterials, Natural Raw Materials, Dielectric Ceramics, Advanced Ceramic Characterization and Applications of nano ceramics in Electric and Radiation fields. She is a member in many local and international research projects. She is a reviewer for many international journals. She has an awarded patents, published book entitled in “Phosphate- containing wastes for fine ceramics production”, and many research papers in international journals. She is a member in the Social Committee of the National Research Center.

Dielectric Evaluation of the fabricated Alumina Composites doped with Different Oxides for Electrical Insulation Applications

Dielectric ceramics are the basic building unit of every electronic device such as ceramic capacitors and microwave dielectric components. This study focused on the fabrication of ceramic composites with good dielectric characteristics and the improvement of the physico- mechanical properties of alumina insulating bodies by doping with different oxides and rare earth elements. Doping the ceramic materials such as alumina with different percentages of ZnO, MgO, spodumene or small content of rare earth elements such as Gd2O3 and Nb2O3 was successful in the enhancement of densification behavior, mechanical properties and the improvement of dielectric characteristics of the fabricated alumina bodies. Such fabricated ceramic materials are promising for utilization in the high voltage transmission line insulator, and for high power circuits that operate at high speed. The doping of alumina with MgO improved the breakdown voltage and resistivity. Further improvement was obtained by co-doping the alumina with Gd2O3 and MgO. The lower the Gd2O3 mass % addition, the better the dielectric properties. Doping the alumina bodies with 7 % ZnO improved both the bodies’ densification behavior and mechanical properties. The results positively confirmed the impact of the addition of ZnO on the resistivity of the samples. Doping the alumina bodies with spodumene and Nb2O3 was successful which promoted the early sintering of the composites at lower firing temperature in addition to the improvement in the dielectric characteristics. The best electrical properties were obtained which are considered satisfying to meet the requirements of high voltage insulating materials.

Dr. Rim Gasmi is an academic researcher and innovation leader specializing in Artificial Intelligence, intelligent systems, and science-based entrepreneurship. She currently serves as Director of the Artificial Intelligence House and the Institute of Sciences at the University Center of Illizi, Algeria, where she plays a key role in developing research, training, and innovation strategies.

Her work focuses on applying AI and IoT technologies to solve real-world challenges, particularly in desert and resource-constrained environments. She has led and contributed to several interdisciplinary research projects addressing road safety, animal vehicle collision prevention, smart monitoring systems, and sustainable development. One of her flagship initiatives involves the design of an AI-based smart collar system aimed at reducing traffic accidents involving animals in desert regions.

Dr. Gasmi is actively involved in empowering young researchers and students by integrating research, entrepreneurship, and responsible innovation into academic programs. She has supervised student-led research projects that resulted in scientific publications and international conference participation.

She was selected as an Emerging Leader in STEM by the International Institute of Education (USA) through the TechWomen program, recognizing her contributions to innovation, leadership, and technology-driven social impact. Her academic interests include ethical AI, inclusive innovation, and the translation of scientific research into sustainable entrepreneurial solutions. Through her work, she seeks to strengthen the role of science and technology in achieving sustainable development goals in Africa and beyond.

Artificial Intelligence as a Catalyst for Science-Based Innovation and Entrepreneurship in Desert Regions

Desert regions face complex and persistent challenges, including road safety risks, limited infrastructure, environmental constraints, and restricted access to technological solutions. Traditional approaches often fail to address these issues effectively due to harsh climatic conditions and vast geographical areas. This contribution explores how Artificial Intelligence (AI) can serve as a powerful catalyst for science-based innovation and entrepreneurship in desert environments.

The proposed approach emphasizes transforming academic research into practical, scalable, and impact-driven solutions by integrating AI, Internet of Things (IoT), and data analytics. AI-driven systems enable predictive analysis, real-time monitoring, and intelligent decision-making, allowing the development of innovative applications in areas such as road safety, smart agriculture, environmental monitoring, and resource management. These technologies not only address critical societal problems but also create new opportunities for entrepreneurship and job creation.

Special attention is given to the role of young researchers and emerging innovators in bridging the gap between scientific research and market-oriented solutions. By fostering collaboration between universities, startups, policymakers, and industry stakeholders, desert regions can become living laboratories for sustainable and inclusive innovation.

This work highlights that desert areas should not be viewed solely as vulnerable or disadvantaged zones, but rather as strategic spaces for scientific experimentation and technological entrepreneurship. Empowering science-based innovation in these regions contributes directly to sustainable development, economic resilience, and societal well-being, while reinforcing the importance of ethical and responsible AI for long-term impact.

Mona Samir Hassan is an Assistant Researcher at the Nanotechnology Research Center (NTRC), The British University in Egypt, with a research background in Organic Chemistry and Nanotechnology. She holds a master's degree in Organic Chemistry and has multidisciplinary research experience in the design, synthesis, functionalization, and characterization of advanced functional materials for Solar Cell, biosensing and biomedical applications.

Her research interests focus on the development of functional materials with tailored chemical and physicochemical properties for advanced applications. Her research experience includes working with graphene-based materials, Metal Oxides, gold nanoparticles, carbon-based nanomaterials, and covalent organic frameworks (COFs), with a particular interest in their integration into functional sensing platforms.

Mona has practical experience in the synthesis and characterization of nanostructured and functional materials, as well as in the investigation of their structure–property relationships. Her technical expertise includes Fourier-transform infrared spectroscopy (FTIR), thermal analysis, and various lab techniques for material preparation, purification, and characterization.

Her work reflects a strong interest in interdisciplinary research, combining organic chemistry, materials science, and nanotechnology to develop functional materials with potential applications in advanced biosensing and biomedical technologies.

Advanced Functional Materials for Biosensing and Biomedical Applications

Advanced functional materials have emerged as powerful tools for the development of innovative biosensing technologies and biomedical applications. Their unique physicochemical properties, tunable surface chemistry, and ability to interact with biological molecules make them highly attractive for the design of sensitive, selective, and efficient sensing platforms.

This presentation will highlight the development and application of functional materials for biosensing, with a focus on the relationship between material structure, surface properties, and sensing performance. Different classes of functional materials, including graphene-based materials, gold nanoparticles, and covalent organic frameworks (COFs), will be discussed as versatile platforms for the development of advanced sensing systems. The role of material design, synthesis, functionalization, and characterization in tailoring the properties of these materials for specific applications will also be addressed.

Particular emphasis will be placed on how the integration of functional materials with appropriate recognition and sensing strategies can enhance analytical performance and enable the detection of biologically relevant targets. The potential of these materials in biomedical applications, including the development of innovative diagnostic and monitoring platforms, will also be explored.

Overall, the presentation aims to provide an overview of the opportunities and challenges associated with the development of functional materials for biosensing and biomedical applications, highlighting how advances in materials chemistry can contribute to the design of next-generation sensing platforms with improved sensitivity, selectivity, and practical applicability.

Solaf Chenini, University Center of Illizi, computer science graduate. I am a productive, disciplined, and hardworking Computer Science professional specializing in Artificial Intelligence. Recognized for consistently performing at the top tier of my academic promotion (Class A) while balancing rigorous external responsibilities, I have demonstrated a strong ability to thrive under pressure and multitask effectively. With two years of hands-on experience as a Commercial Agent at a major national enterprise (Sonelgaz) alongside a background in teaching, I bring a well-rounded work ethic, advanced technical expertise, and a proven capacity to excel in both corporate and academic roles.

Hand gesture and sign language recognition for Targui Language based on deep learning

To reduce communication barriers between the deaf population and those who do not use sign language, Targuia Sign Language was created.This research presents the deep learning (DL) framework for Targuia Sign Language recognition. Such a Convolutional Neural Network (CNN) is used to extract the strong and unique spatial features from the sign language image and thus different hand gestures can be classified correctly. The proposed system is trained and tested on a specific dataset which contains several examples of images of Targuia Sign Language. We extensively evaluate the model and show that it achieves high recognition accuracy for multiple gesture classes and different users. Furthermore, the proposed method can be extended to be used for real-time sign language detection in conjunction with a camera-based input device.

Dr. Ghada Mohamed Taha Ibrahim is an Associate Professor at the Textile Research and Technology Institute, National Research Centre (NRC), Cairo, Egypt. She received her B.Sc., M.Sc., and Ph.D. in Organic Chemistry from Ain Shams University. Her Ph.D. research focused on the development of multifunctional finishing technologies for cotton fabrics.

Her research focuses on nanomaterials, smart and functional textiles, green textile finishing, hydrogels, sustainable packaging materials, biomedical applications, and wastewater treatment using advanced adsorbent materials. She has extensive experience in the synthesis and characterization of polymeric and nanostructured materials for environmental and healthcare applications.

Dr. Taha has authored more than 35 peer-reviewed publications in high-impact international journals and has an **h-index of 18**, with over **860 citations**. She has participated in numerous national and international research projects, serves as a reviewer for several leading scientific journals, and is actively involved in scientific conferences, research collaboration, and laboratory quality management according to ISO/IEC 17025 standards.

Controlled Delivery of Trans- cinnamaldehyde through Chitosan/Alginate Nano capsules for Active Food Packaging Textiles

The practical application of natural antimicrobial compounds in active food packaging remains challenging because of their high volatility, susceptibility to degradation, and rapid release, which limit their long-term effectiveness. In this study, biodegradable chitosan/alginate Nano capsules were developed as a bio-based delivery platform for the controlled release of trans-cinnamaldehyde (CINA) and subsequently applied to cotton fabrics to produce multifunctional active food packaging textiles. The Nano capsules were successfully fabricated through polyelectrolyte complex coacervation and immobilized onto cotton substrates using a conventional textile finishing process.

The physicochemical characterization confirmed the successful formation of the Nano capsules, efficient encapsulation of trans-cinnamaldehyde, and its homogeneous distribution on the cotton fiber surface. Encapsulation significantly improved the structural organization of the Nano carrier system while maintaining good colloidal stability and high encapsulation performance. Moreover, the developed Nano capsules provided sustained release of the encapsulated bioactive compound, demonstrating a controlled release behavior that is expected to prolong antimicrobial functionality compared with the free compound.

The functionalized cotton fabrics exhibited pronounced broad-spectrum antimicrobial activity against representative foodborne pathogens together with enhanced antioxidant performance, confirming that Nano encapsulation effectively preserved the biological activity of trans-cinnamaldehyde while improving its stability and availability. The optimized formulation achieved superior antimicrobial efficacy using a relatively low concentration of the active compound, highlighting the efficiency of the developed delivery system.

This study demonstrates that chitosan/alginate Nano capsules provide an effective and environmentally friendly platform for the controlled delivery of volatile natural antimicrobials. The developed bioactive textile system combines sustained release, enhanced antimicrobial protection, and antioxidant functionality, offering a promising strategy for next-generation active food packaging applications aimed at improving food safety and extending product shelf life.

Sarah Hesham Mohamed Rashed, a Ph.D. researcher at E-JUST, SRTA CITY specializing in green energy and sustainable development. I am Assistant Lecturer at Borg El-Arab Technological University. I am an instructor in AI and IoT sensors of wearable devices. I am also the focal point for the Rally Entrepreneurship Community and was honored as the Best Multidisciplinary Researcher at the 2025 Global Research Conference for "AI for a Better Future." Women Award in Innovation and Technology" in DUBAI 2025.

As an international keynote speaker and mentor in nano science and technology and ICT, I have contributed to global forums like COP29 and COP 30, and specialize in leveraging material science, AI-driven analytics, automation, and connected intelligence to optimize energy systems. By bridging the gap between technical research, technology and entrepreneurial leadership, I aim to empower the next generation of innovators to drive global sustainability goals.

Dyeing to Be Clean: How Double Layered Hydroxides are Revolutionizing Industrial Wastewater Treatment

Double layered hydroxides sodium dodecyl sulphate (LDH-SDS) and Graphene double layered hydroxides (GO-LDH) adsorbents are synthesized by reconstruction and sonochemical methods, respectively, to adsorb Coomassie brilliant blue G-250 (CBB) from an aqueous solution. The adsorbents are investigated by XRD, SEM, EDS, FTIR, Raman spectra, TGA and Zeta-sizer. According to the characterization results, GO-LDH has better properties than LDH-SDS, resulting in greater CBB adsorption. The batch experiment is done under various parameters such as: contact time, initial dye concentration, adsorbent dose, PH, temperature, and ionic strength effect. It’s found that 5 mg of GO-LDH increased the adsorption capacity to 1443 mg/g for 5 minutes at higher initial dye concentration, compared to LDH-SDS of 60 mg at 85 mg for 1 hour at lower initial dye concentration. GO-LDH has a pseudo-second-order initial adsorption rate that is 7 times that of LDH-SDS and has higher boundary layer diffusion, according to the intra-particle diffusion model, than LDH-SDS. Furthermore, GO-LDH obeys the Freundlich isotherm with (1/n) 0.406, whilst LDH-SDS obeys the Langmuir isotherm with (qm) 60 mg/g. The thermodynamic parameters show that the adsorption is restricted, exothermic, and spontaneous. Moreover, GO-LDH has a more effective recycling process than LDH-SDS using the memory effect method.

Session 3 Materials-Engineering:

Wednesday, 9th September, 11.00 am-1.00 pm (2 hours), National Research Center.

Professor Hanan Basioni Ahmed is a full professor at the Faculty of Science, Helwan University, Egypt. She is also an Adjunct Professor at Faculty of Engineering, Chulalongkorn University, Bangkok, Thailand. Professor Hanan Basioni is a member of the National Committee of Pure and Applied Chemistry (2025-2028) that contributes to scientific policy development, interdisciplinary research and the promotion of applied scientific innovation.

Professor Hanan Basioni is a member of several scientific societies, such as the Egyptian Chemical Society, the Egyptian Polymer Society, the Egyptian Society for Textile Industries, and is also a member of the Association for Women in Science in Developing Countries.

In recognition of her scientific excellence, Professor Hanan has received several distinguished awards, including the Egyptian State Encouragement Award (2021), Helwan University Award for Scientific Excellence (2022), and the Helwan University Appreciation Award (2024). Professor Hanan's name was included in Stanford's list of the top 2% of scientists for 2021, 2022, 2023, 2024, and 2025.

She has authored over 75 high-quality publications in prestigious international journals with strong impact factors, has contributed 6 chapters to international books, receiving more than 4,300 citations and achieving H-index of 42. Professor Hanan supervised more than 16 Master's and Doctoral theses at faculty of science-Helwan University.

The field of interest for Professor Hanan Basioni is focused on obeying green chemistry concepts for designing various types of nanomaterials to be applicable in biomedical technologies, environmental sustainability, and various industrial purposes.

Green fabricated Nanomaterials for Pharmaceutical and Biomedical purposes

The growing demand for sustainable and biocompatible materials has driven significant interest in green-fabricated nanomaterials for pharmaceutical and biomedical applications. Green chemistry-based fabrication approaches utilize natural extracts, biopolymers, microorganisms, and environmentally benign reagents to produce nanomaterials with reduced toxicity, lower energy consumption, and minimal environmental impact compared to conventional synthetic methods. These eco-friendly nanostructures exhibit enhanced physicochemical properties, including controlled particle size, improved surface functionality, and superior biological compatibility, making them highly suitable for drug delivery, diagnostics, antimicrobial therapy, tissue engineering, and bio-sensing applications. Green-fabricated nanomaterials have demonstrated remarkable therapeutic efficiency through improved drug solubility, targeted delivery, and controlled release profiles, leading to enhanced bioavailability and reduced side effects. Moreover, their inherent biological activity derived from natural reducing and stabilizing agents often contributes to synergistic pharmacological effects, particularly in antimicrobial, antioxidant, and anticancer therapies. Despite their promising potential, challenges remain in large-scale production, reproducibility, and regulatory standardization. Addressing these limitations will be essential to translate green nanotechnology from laboratory research to clinical and industrial implementation. Overall, green-fabricated nanomaterials represent a sustainable and innovative platform for advancing pharmaceutical and biomedical technologies, aligning therapeutic development with environmental responsibility and safer healthcare solutions.

Dr. Moshera Samy Abdelaziz Youssef, was born Cairo, Egypt in 1984. She holds a PhD by the Faculty of Science, Ain shams University (2019). She is a Associate Prof. Dr. of Polymers technology at National Research Centre (NRC), Giza, Egypt at 14/04/2025. She has got National Research Center Prize of for the best Master thesis Egypt for 2013. She has been awarded the BSc degree at the Faculty of Science, Cairo University, Egypt (2006). She graduates courses as a partial fulfilment of the Master degree of Science, Organic Chemistry, Department of Chemistry, Faculty of Science, Cairo University, Egypt (2007). She has participated in many national and international conferences. She has participated in six national and international research projects.

She is a Member of The Egyptian society of polymers and pigments and young researchers in NRC, Giza, Egypt. She held postdoctoral positions in Deusto Institute of Technology (DeustoTech), Bilbao, Spain (2022) for six months. She focuses her research on Synthesis and characterization of biodegrade Polymers by using blending and casting solvent method and studying their applications in biosensors. This would help in understanding the effect of different methods governing the successful

mechanical properties and improvement of their toughness of the prepared biodegradable Polymers for efficient delivery of anticancer drugs. Her fellowship is funded by pregame Science by Women. Her fellowship is sponsored by Diputación Foral de Vizcaya. Now, I'm an Associate Prof. Dr. of Polymers technology at National Research Centre (NRC), Giza, Egypt at 14/04/2025.

In vitro release and cytotoxicity activity of 5-fluorouracil entrapped polycaprolactone nanoparticles

In this study, 5-fluorouracil (5-FU) entrapped polycaprolactone nanoparticles (5-FU- PCNPs) have been prepared using double emulsion method. The different factors were examined for assembly to arrive at the best effective formulation of 5-FU-PCNPs formulation for 5-FU–PCNPs, as polymer concentration, stabilizer concentration. The encapsulation efficiency of PCNPs was in the range of 18.8–45.4%. The prepared nanoparticles showed the spherical shape having an average size of 183–675.5 nm, whereas TEM exhibited the prepared nanoparticles have a spherical shape. FTIR, XRPD, confirmed successful insertion of drug in prepared PCNPs. In vitro release of 5-FU from selected formulations showed sustained release from the nanoparticles where slower release was observed when lower PVA concentration was used. Anticancer activity was examined against cell culture for HCT-116 (human colorectal carcinoma), MCF-7(human breast adenocarcinoma), HepG2 (human hepatocellular carcinoma) and A549 (human lung carcinoma) for six formulations 5-FU–PCNPs nanoparticles. The in vitro cytotoxic activity of the prepared formulations was tested showing that these formulations appeared as promising active anticancer formulations.

Dr. Rasha A. Baseer is an Associate Professor in the Department of Polymers and Pigments Technology at the National Research Centre (NRC), Egypt. She earned her B.Sc. degree from the

Faculty of Science, Benha University, Egypt, and her Ph.D. in Organic Chemistry from the Faculty of Science, Ain Shams University, Egypt.

Dr. Baseer's research expertise encompasses polymer chemistry and its biomedical, electrical, and environmental applications. Her work focuses on the design and development of advanced polymeric materials for biomedical devices, conductive composites, energy storage, and water treatment. She also has extensive experience in the chemical recycling of plastic waste and its valorization for environmental applications, particularly water purification.

In 2019, Dr. Baseer completed specialized training in the pulp and paper industry based on non-wood fibers at the Chinese National Pulp and Paper Research Institute, People's Republic of China.

Dr. Baseer has authored numerous peer-reviewed publications in leading international journals and is an inventor on two awarded Egyptian patents: "Method for Open Cyclic Polymerization of Ethylene Carbonate" (2022) and "Polymeric Derivative of (N-Benzyl-3-acetyl Indole)" (2025). She is also the inventor of two pending Egyptian patent applications: EG/P/2022/1313, New Formula of Gelatin-co-polyacrylamide/Nano Zinc Oxide Hydrogel Based on Cotton Gauze for Wound Healing, and 429/2018, Synthesis of New Poly(indole/thiazole) Derivatives.

Throughout her career, Dr. Baseer has participated in numerous national and international research projects, serving both as a principal investigator and as a research team member. She is an active member of the Egyptian Society for Polymer Science and Technology and the Organization for Women in Science for the Developing World (OWSD). In addition, she serves as a reviewer for approximately 40 manuscripts across about 15 international scientific journals published by major academic publishers.

Enhanced electrochemical and optical properties of new conjugated n-p type polymers

This research work is intended for the novel synthesis and study of structural, thermal, optical, electrical, and dielectric properties of semiconductor polyindole fused thiazolo-thiadiazole (PAITTD) and its N-substituted indole derivatives with a side chain containing chlorobenzoyl (PACBITTD) and bromobenzene sulphonyl (PABBSITTD). The characterization techniques such as H1NMR, FTIR, SEM, and TGA/DSC ensured the structural and thermal properties of the as-synthesized polymers. The optical band gap as obtained from the absorbance study was found to be in the range of 1.5 eV to 2.58 Ev. Moreover, the ground-state geometries and the absorption wavelengths for the studied compounds were investigated by DFT and TD-DFT, where the theoretical results show coincidence with these of the experimental where the calculated (λmax) values are approximately near to the experimental (λmax) values for the PAITTD, PACBITTID, and PABBSITTD. The impedance ac conductivity and dielectric properties of the PAITTD, PACBITTID, and PABBSITTD were performed in the frequency ranges from 20 Hz–10 MHz at room temperature, where their values were 3.0×10^{-7}, 8.7×10^{-7}, and 2.6×10^{-6}, respectively. Based on the optical and electrical properties, synthetic PAITTD, PACBITTID, and PABBSITTD are promising materials in the design of optoelectronic semiconductor devices.

Dr. Eman Mahmoud obtained her BSc in (Physics and Chemistry), MSC in (Biophysics), and PhD in (Biophysics) from the Faculty of Science, Ain Shams University, in 2006, 2012, and 2016, respectively. She is currently associate Professor of Ceramics Chemistry and Materials Technology in the Refractories, Ceramics and Building Materials Department, Advanced Materials Technology and Mineral Resources Research Institute, National Research Centre. Dr. Mahmoud's research interests include Ceramic Science and Technology, Biomaterials, Nanomaterials, Natural Row Materials, Advanced Bioceramics, Blood analysis and Biochemistry evaluations, Tissue engineering, In-Vitro and In-Vivo evaluations and characterizations of biomaterials, bone graft implant, bone substitutions Materials. Dr. Mahmoud received the Second Cairo Innovation Exhibition Award in 2015. Dr. Mahmoud has published over 17 articles in international journals with a Scopus H-index of 10. She was also a member of several research projects international, national, and a national need project including an industrial project under Title Maximizing the utilization of eggshell wastes resulted from sterilized egg industry for industrialization of high-economic-value products (2023-2024), also Reviewer activity for several international journals.

Recent trends in Advanced Bioceramics for orthopedic Applications

Every year, millions of patients require bone replacement or grafting to replace bone mass lost due to ageing, bone defects, and traumas. There is an increasing need for sustainable prosthetic devices and associated medical systems due to the world's population's longer life expectancy. Utilization of advanced bioceramics in tissue engineering has emerged as a promising approach to treat the injurie or loss bone. Great efforts have been made to improve bone ingrowth. Bioceramics are considered among the essential materials employed in the biomedical field as they were created to be suitable for connection with living tissues. Their biological attitude is based on their chemical constituents and physical features. They are resorted to instead of metallic biomaterials due to their unique properties, such as biocompatible, corrosion resistance and bioactivity. Injectable hydrogel has recently gained greatly of attention in the field of reconstructive bone surgery. Using injectable hydrogels offers several advantages, including providing an easy strategy for healing irregular and complex bone defect areas with an injection needle, as well as reducing recovery time by avoiding away with open surgery procedures. Our study's goal is to shed light on recent trends in advanced bioceramics for orthopedic Applications, properties, Preparation, Characterization, In-Vitro and In-Vivo evaluations. In addition, it discusses and summarizes the considerable advancements in the use of advanced ceramic materials for bone replacement as the perfect choice for orthopedic and maxillofacial applications. Additionally, it highlights of traditional and modern research. It defines various ceramic materials, their bioactivity, clinical applications, advantages, limitations, fabrication techniques, mechanical and physical characteristics, and the criteria for selecting long-lasting implant devices for advanced ceramic materials.

Dr. Heba-tollah Sweelam is an Associate Professor in Chemistry of Natural Products, she works at the chemistry of natural compounds department, pharmaceutical and drug industries research institute, National Research Centre, Egypt. She earned her Ph.D. from the botany and microbiology department, college of science, Cairo University. Her research is focused on the isolation and identification of bioactive compounds from plant origin and their potential health promoting properties, as well as practicing different tissue culture techniques to increase the content of bioactive compounds in the regenerated plants.
Her research interest includes metabolomics, tissue culture and antimicrobial activity. She has experience in modern analytical techniques, including Nuclear magnetic resonance spectroscopy (NMR), liquid chromatography tandem mass spectrometry (LC MS/MS), and molecular networking for the identification and characterization of natural products. She actively collaborates with multidisciplinary research groups and has published more than thirteen scientific papers along with book chapters; her research integrates phytochemical analysis, chromatography and spectroscopy. She has participated in collaborative scientific projects related to plant based bioactive compounds. Also, she is a member of many scientific research projects. She is also a member of many scientific societies such as the society for women in science in the developing countries, the Arab society for medical research and the Egyptian chemical society along with many more. She has contributed scientifically in peer reviewed scientific papers. In addition to her research activities, she assists in undergraduate laboratory teaching and mentor students in phytochemistry techniques, and participates in many workshops focused on advanced laboratory instrumentation.

In Vitro Culture as a Sustainable Platform for the Production of Pharmaceutically Important Natural Products from Higher Plants

Higher plants, especially wild ones, are considered a source of many important medicinal compounds, fragrances, and dyes. Plants undergo a variety of stresses and harsh conditions to help them flourish and cope with such situations, so they produce a wide range of secondary metabolites that are essential for defense against biotic and abiotic stressors. Notwithstanding their importance and due to their numerous industrial uses, in addition to the limited natural yield of these bioactive chemicals, genetic variability and environmental factors make their extraction and use extremely difficult. Plant tissue culture (cultivation of a small plant part for regeneration of a new plant) has become a competitive substitute in recent years for the regulated synthesis of secondary metabolites through a number of productivity-boosting techniques, such as elicitation and precursor feeding. The main benefit of this technology is to provide a reliable and continuous source of plant pharmaceuticals that could be utilized for the large-scale cultivation of plant cells, from which these metabolites can be extracted. Also, it can be used as a substitute for wild plants due to many reasons, including their slow growth rate, low concentrations of active compounds, and the requirement for biotic and biotic stress to induce biosynthesis of the secondary metabolites. To increase the production of bioactive compounds and boost their synthesis, numerous tactics need to be used including metabolic engineering and up-scaling using effective bioreactor approach. This work presents an updated overview of the use of in-vitro culture for the production of pharmaceutically important plant metabolites.

Dr. Mona Sayed Ahmed, Associate Professor of Ceramic Chemistry and Materials Technology in the Refractories, Ceramics, and Building Materials Department, Advanced Materials Technology and Mineral Resources Research Institute, National Research Centre (NRC), Egypt. She has published 26 research articles in international journals, one book, one review article, and is the inventor of one granted patent. Participated as a researcher in many national and international research projects. Former Technical Examiner at the Egyptian Patent Office. Member of the Technology Transfer Office (TICO) at the National Research Centre (2020–2024). Member of the Office of Technology Commercialization (OTC) at the National Research Centre. Throughout her academic career, she has participated in several international fellowships and scientific visits to Germany, Italy, Poland, and China.

Bioactive Calcium Phosphate - Silicate Composites: Synthesis, Characterization and In-vitro Assessment

Bioceramic materials have attracted considerable attention in regenerative medicine due to their biocompatibility and bioactivity. Among these materials, calcium phosphate ceramics like tricalcium phosphate and hydroxyapatite closely resemble the mineral composition of natural bones. However, their limited mechanical properties and slow degradation rates may restrict their applied in load-bearing application. On the other hand, silicates based materials possess high bioactivity and biodegradability properties. Accordingly, this work focus on development and characterization of bioceramic composites combining calcium phosphate and silicate phases to achieve enhanced in biological and physical properties. Furthermore, the controlled release of calcium, phosphate, and silicon ions contributes to cellular proliferation, differentiation and supporting accelerated bone healing. Preparation methods, including nano-calcium phosphate and silicates powders by precipitation and sol-gel techniques were achieved, respectively. The formed composites powder was pressed into pellets to obtain bioceramic samples and sintering at their optimum temperature. The results demonstrated that Calcium phosphate-silicate bioceramic composite exhibited excellent in vitro bioactivity following simulated body fluid immersion. Cytotoxicity assessment revealed no significant toxic effect. Cell viability studies showed high cell proliferation and viability. The obtained results suggest that the prepared composites have significant potential to be applied as promising biomaterials for bone tissue engineering applications.

Dr. Samaa M. Faramawy, Associate professor and postdoc researcher at the National Institute of Standards (NIS). My current research interests are, Experienced researcher in radiometric metrology in fields of solar-radiation measurements, UV-Radiation measurements, Phototherapy, Ultraviolet Therapy measurements, and UV-INDEX sensation. I am also Competent in light metrology science (electromagnetic radiation), General principles of measurement and instrumentation, Sensors and sensor systems, Specific applications in measurement and instrumentation, Applications of sensors and instrumentation.

Metrology and Traceability: Role as a Precondition for Sustainability

This analysis examines the often-overlooked but strictly necessary role of metrology in the successful execution of the Sustainable Development Goals. We argue that because you cannot manage what you cannot measure, traceable, accurate measurement is a precondition for a sustainable future by 2030. The work presented here focuses on technical applications where standard and certified measurements directly facilitate the realization of specific SDGs. (1) Solar Resource Assessment: The presentation demonstrates how standard and traceable measurements of solar irradiance (W/m^2) are critical for reducing technical uncertainty. This precision facilitates data-driven investment decisions, precise performance modeling, and maximized efficiency for solar generation facilities, thereby directly enabling Goal 7 (Affordable and Clean Energy) and Goal 13 (Climate Action). (2) UVC Disinfection Validation: The metrological calibration of UVC sensors ($\mu W/cm^2$) provides the requisite data to validate adequate radiometric dosing. This validation is essential for confirmed pathogen and microorganism inactivation without wasteful over-dosing, directly supporting Goal 6 (Clean Water and Sanitation) and Goal 3 (Good Health and Well-being). (3) Medical Phototherapy Precision: The presentation will show the essential need for accredited calibration of phototherapy devices and testing different sources. We ensure specific spectral irradiances (($\mu W/cm^2/nm$) for neonatal applications, ($\mu W/cm^2$) for dermatological applications), preventing dangerous dosage errors and serving Goal 3 (Neonatal and Public Health) and Goal 9 (Industry, Innovation, and Infrastructure) through standardized safety protocols.

Ultimately, metrology provides the required bridge from scientific principles to verifiable field impact, serving as the guarantor of human safety in purification and phototherapy, critical water purity validation, the structural reliability of sustainable energy infrastructures, and other essential applications.

Dr. Mona Moaness Abdel-Moneam Mohamed is an Associate Professor of Biomaterials at Refractories, Ceramics and Building Materials Department, Advanced Materials Technology and Mineral Resources Research Institute, National Research Centre (NRC) in Cairo, Egypt, specializing in nanotechnology, advanced drug delivery systems, and regenerative medicine. With a Ph.D. in Biochemistry from Cairo University, her pioneering research centers on the fabrication of bioresorbable electrospun nanofibers, 3D-printed scaffolds, and smart pH-sensitive hydrogels designed to improve bone regeneration and accelerate skin wound healing. Her impactful work has earned her the 2024 NRC President's Award for Best Applicable Research and a co-invented patent for skin regenerative patches. In addition to her extensive record of peer-reviewed publications, international grant collaborations, and journal reviewing duties, Dr. Mona is deeply committed to academic mentorship—supervising graduate theses and student training programs successfully

3D-Printed Reinforced Scaffolds for Bone Regeneration and Controlled Drug Delivery

The development of multifunctional scaffolds capable of promoting bone regeneration while preventing implant-associated infections remains a major challenge in bone tissue engineering. In this study, novel 3D-printed sodium alginate/poly (vinyl alcohol) (SA/PVA) scaffolds reinforced with amoxicillin-loaded copper/cerium co-doped 58S bioactive glass nanoparticles (BG-Cu/Ce-Amox) were developed for simultaneous bone regeneration, infection control, and controlled drug delivery. The bioactive glass nanoparticles were synthesized using a sol-gel method and incorporated into the SA/PVA matrix through 3D printing, producing highly porous scaffolds with interconnected architectures suitable for bone tissue growth. Morphological and elemental characterization confirmed the successful incorporation of Cu/Ce-doped bioactive glass and amoxicillin within the polymeric network. Scanning electron microscopy revealed well-defined porous structures and enhanced surface roughness after nanoparticle addition. In vitro bioactivity evaluation in simulated body fluid demonstrated superior apatite formation and calcium-phosphate mineralization compared with pristine SA/PVA scaffolds. The drug-loaded scaffolds retained excellent bioactivity while exhibiting elemental signatures associated with amoxicillin incorporation. Biological assessment using MG-63 osteoblast-like cells showed significantly enhanced cell proliferation and alkaline phosphatase activity after 3 days, indicating improved osteogenic

potential. Antibacterial evaluation against Gram-positive and Gram-negative bacteria demonstrated broad-spectrum antimicrobial activity. Drug release studies in PBS (pH 7.4) revealed sustained amoxicillin release for up to 7 days, enabling prolonged local antibiotic delivery and reducing the need for repeated administration. Overall, these multifunctional 3D-printed scaffolds integrate bioactivity, osteogenic stimulation, controlled antibiotic release, and antibacterial performance, making them promising candidates for bone regeneration applications with enhanced resistance to implant-associated infections.

Hadeer Mamdouh Ahmed Eldeeb, is Egyptian Teaching Assistant at the Organic Conservation Department, Faculty of Archaeology, Cairo University. I hold a Master degree (2022) with a focus on eco-friendly materials for photograph conservation, along with an Excellent grade Bachelorâ€™s degree (2015). My academic portfolio includes published research on fungal effects and private photographic collection conservation. I possess strong technical and practical skills, including proficiency in photography, paper conservation, AutoCAD (2D), Photoshop, and digital transformation. I am a highly adaptable and collaborative professional with excellent communication, teaching, and research abilities.

Novel Nanostructured Agarose Gels for the Non-Invasive Cleaning and Protection of Historic Tintype Photographs

Tintype photographs serve as vital cultural and historical artifacts from the 19th century; however, their long-term preservation is frequently threatened by surface contaminants such as soot and dust. Due to their thin, sheet-like structure, conventional cleaning methods often induce scratching or irreversible damage to the delicate image layer. This study evaluated the cleaning efficacy of alternative approaches, contrasting traditional swab cleaning with agarose gel and nanomaterial-functionalized agarose composite gels incorporating zinc oxide (ZnO), carbon dots (CDots), and graphitic carbon nitride (g-C3N4).The synthesized nanomaterials and composite gels were comprehensively characterized using TEM, DLS, zeta potential, UV-vis, and ATR-FTIR spectroscopy. Cleaning performance was assessed through portable digital microscopy, colorimetric analysis (Î”E), and ATR-FTIR, both before and after treatment, as well as following artificial light aging. The nanomaterial-functionalized gels demonstrated superior cleaning efficiency and protective capabilities compared to traditional methods. Specifically, agarose gels loaded with ZnO (Î”E=3.35), CDots (Î”E=2.7), and g-C3N4 (Î”E=3.09) effectively removed dust without causing perceptible color shifts, while providing an added shield against UV exposure. For soot removal, the agarose + ZnO (Î”E=5.84) and agarose + CDots (Î”E=7.73) formulations achieved the highest performance. Crucially, post-treatment ATR-FTIR analysis confirmed no chemical alterations to the tintype layers, validating these functionalized gels as safe, non-invasive solutions for heritage conservation.

Session 4 Materials-Engineering:

Wednesday, 9th September, 1.00-2.30 (1.30 h), National Research Center.

Prof. Dr. Marwa I Wahba, Professor at the Department of Chemistry of Natural and Microbial Products, and at the Advanced Materials and Nanotechnology group, Centre of Scientific Excellence, National Research Centre, Giza, Egypt. Interested in research on biopolymers and their exploitation as immobilization matrices & packaging materials, and also interested in altered methods that can be adopted to promote enzymes stability.

Proteins as grafting materials for the fabrication of covalent immobilizers

Covalent immobilizers are critical for the industrial exploitation of enzymes as they allow enzymes reuse and promote their stability. Nonetheless, covalent immobilizers are usually fabricated via grafting with synthetic polymers, such as polyethylene-imine (PEI), which is expensive and also toxic. Accordingly, safe and cost-effective grafting materials should substitute the PEI. Proteins, such as egg white protein and whey protein isolate were successfully utilized as PEI substitutes. These proteins were utilized to graft altered biopolymers, such as carrageenan and gellan-gum. The grafted protein then allowed the binding of the covalently reactive glutaraldehyde to the immobilizer matrix, and thus, the covalent immobilizers were attained. These covalent immobilizers successfully immobilized altered enzymes, such as β-galactosidase and cellulase while providing them with excellent reusability and escalated stability.

Dr. Marwa Hany, Assistant Professor of Pharmaceutical and Medicinal chemistry at the German University in Cairo GUC & German International University GIU
She is currently the Program Supervisor of both the faculties of Biotechnology and Pharmaceutical Engineering- German International University (GIU)
Dr Marwa has earned her MSc. and PhD degrees in Pharmaceutical chemistry from the GUC, she is mainly interested in drug design and development of novel anticancer and anti-infectious small organic molecules. in addition, she is also interested in green chemistry applications in pharma industry and applications of Artificial intelligence in drug discovery and has organized some summer workshops in Berlin to introduce to these areas of interest.

Identification of novel pyrazolines and isoxazolines as antileishmanial agents via a potential antifolate mechanism: synthesis, biological evaluation, and in silico studies

Leishmaniasis remains a major neglected tropical disease affecting nearly one million people annually. Due to toxicity and resistance to current therapies, the folate/pterin salvage pathway represents a rational target for antileishmanial inhibitor design. Novel chalcone-derived pyrazolines and isoxazolines were synthesized and evaluated in vitro against Leishmania major. Six compounds (1b-e, 1g, and 1i) showed higher potency than miltefosine. Specifically, the N-formylated pyrazoline derivatives 1g and 1c exhibited exceptional antileishmanial potencies, with IC50 values of 1.41 ÂµM and 1.54 ÂµM against promastigotes as well as 1.71 ÂµM and 1.94 ÂµM against amastigotes, respectively. This represents a significant potency improvement (up to 5.5-fold) over miltefosine (IC50 = 7.83 and 8.07 ÂµM, respectively). The antileishmanial activity was reversed with folic and folinic acid supplementation, supporting an overall antifolate mechanism. Molecular docking and 100 ns molecular dynamics simulations were conducted within the binding pocket of the essential Leishmania major pteridine reductase 1 enzyme (Lm-PTR1; PDB ID: 7PXX), demonstrating stable ligand-protein complexes for the most active candidates. In addition, molecular docking and 100 ns MD simulations of the S-enantiomer of compound 1g against a validated AlphaFold2 model of Lm-DHFR revealed favorable binding scores and stable cofactor-pocket interactions, suggesting a predicted dual-target inhibitory potential against both Lm-PTR1 and Lm-DHFR. Compounds 1g and 1c showed minimal cytotoxicity against Vero cells, with selectivity indices (SI) of 148.7 and 260.9, respectively, compared to miltefosine (SI = 12.6). Based on its superior potency, high selectivity, and stable target binding in silico, the S-enantiomer of compound 1g represents a promising lead for further antileishmanial development.

DR. Maha Ahmed Ali Ahmed is an Associate Professor of photograph conservation at the Faculty of Archaeology, Cairo University. In 2016, she received her doctorate degree in Conservation from Cairo University. She was a member of a research team in a joint project between Cairo University and University of Catania dedicated to photograph preservation during the period from 2013 2017. She also taught several photograph conservation classes at post graduate level in a joint International Masters Program in Conservation of Antique Photographs and Paper Heritage between Helwan University and University of Catania. She has also held a number of workshops in Egypt, Lebanon, and Morocco with the aim of raising awareness on the significance of photograph heritage in Egypt and the Middle East and their preservation needs. Maha has published several papers discussing different issues related to photograph conservation. Her primary research area is the utilization of advanced analytical techniques and technologies in photograph conservation field. She is an alumna of the Middle East Photograph Preservation Initiative (MEPPI), a strategic multi-year effort funded by the Andrew W. Mellon Foundation and successfully organized by the Arab Image Foundation (AIF), the Art Conservation Department at the University of Delaware, the Metropolitan Museum of Art and the Getty Conservation Institute to build the capacity of individuals and institutions in the care and preservation of photograph collections in the broad Middle East, from North Africa and the Arabian Peninsula through Western Asia. Maha was also a member of the MEPPI Steering Committee to whom the responsibility to drive the initiative into new grounds and provide a road map for the future of MEPPI was entrusted. She participated in several conservation projects related to photograph conservation (e.g., the photo collection at the library of the University of Alexandria).

Mount Arafat Through the Lens of Muhammad Sadiq Bey: Documentation and Analysis of a 19th Century Albumen Print

Muhammad Sadiq Bey, an Egyptian army engineer, surveyor, and treasurer of the Egyptian Hajj caravan, is recognized as the first photographer to document Mecca, Medina, and the Hajj pilgrimage. This study presents the documentation, material characterization, and deterioration assessment of a 19th century albumen print depicting Mount Arafat, photographed by Sadiq Bey. The photograph is mounted on a low-quality secondary support and preserved within a wood frame. Given the inherent instability of albumen prints, a study was conducted to evaluate the photographâ€™s condition and identify the deterioration forms affecting it. The investigation employed visual examination, stereomicroscopy, ultraviolet (UV) and infrared (IR) imaging, scanning electron microscopy coupled with energy-dispersive X-ray spectroscopy (SEM-EDS), pH measurement, Fourier transform infrared spectroscopy (FTIR), and microbiological analysis. The results revealed extensive deterioration, including yellowing and flaking of the albumen binder, fading and discoloration of the silver image, and yellowing and embrittlement of the secondary support. Evidence of insect infestation, including damage associated with Anthrenus pimpinellae larvae, and microbiological activity was also identified. IR imaging revealed evidence of retouching, while SEM examination showed micro-cracking and fungal spores. EDS analysis detected silver sulfide, contributing to image discoloration. The secondary support exhibited an acidic pH of 4.9, while FTIR indicated severe degradation of the albumen binder and oxidative deterioration of the secondary support. Microbiological analysis isolated Aspergillus niger and Emericella nidulans, with A. niger identified as the probable primary contributor to biodeterioration. The findings provide valuable insights into the material composition, deterioration forms, and preservation challenges of this historically significant photographic record of Mount Arafat.

Dr. Sara Sayed Abdel-Haliem Selim is a researcher at the Egyptian Petroleum Research Institute (EPRI), Cairo, Egypt. Her research focuses on the development of advanced materials for environmental remediation, with particular emphasis on adsorption technologies, water and wastewater treatment, and sustainable waste management. Her work includes the design and application of porous materials, nanocomposites, and carbon-based adsorbents for the efficient removal of pollutants from water and industrial effluents.
Her research interests include adsorption science, environmental nanotechnology, oil spill remediation, wastewater purification, nanostructured materials, and the valorization of plastic and industrial wastes into value-added functional materials. She is committed to developing innovative, cost-effective, and environmentally sustainable technologies that support resource recovery and circular economy principles.

Dr. Selim has published extensively in international peer-reviewed journals and has presented her research at numerous national and international scientific conferences. Through multidisciplinary collaborations, she continues to advance sustainable materials and technologies for environmental protection, clean water production, and pollution control.

Plastic Waste to High-Performance Aerogel Sorbents for Sustainable Oil Spill Cleanup

The escalating accumulation of plastic waste and the increasing occurrence of oil spills have emerged as two critical environmental challenges, necessitating innovative and sustainable remediation technologies. This study presents a circular economy strategy for converting discarded plastic waste into ultralight, highly porous aerogel sorbents for efficient oil spill cleanup and wastewater purification. The plastic waste-derived aerogels were fabricated through a facile and environmentally friendly process, producing a three-dimensional interconnected porous network with low density, high porosity, and excellent mechanical stability. Owing to their intrinsic hydrophobicity and superoleophilicity, the aerogels selectively absorb a wide range of petroleum oils and organic solvents while effectively repelling water. Comprehensive structural and physicochemical characterization was performed to correlate the morphology, pore architecture, surface chemistry, and wettability with the sorption performance. Batch sorption experiments demonstrated rapid oil uptake, high absorption capacities, fast sorption kinetics, and excellent selectivity toward various hydrocarbons under different operating conditions. Furthermore, the aerogels exhibited remarkable recyclability, maintaining high sorption efficiency and structural integrity over repeated absorptionâ€“recovery cycles through simple mechanical squeezing or solvent extraction. The interconnected porous framework promoted efficient capillary-driven transport and storage of oils, while the robust structure prevented collapse during repeated use. By transforming persistent plastic waste into high-value functional sorbents, this work simultaneously addresses plastic pollution and oil contamination, providing a sustainable waste-to-resource platform for environmental remediation. The developed plastic waste aerogels offer significant potential for large-scale oil spill response, industrial wastewater treatment, and resource recovery, representing an economically viable and environmentally friendly alternative to conventional oil sorbent materials.

Dr. Heba Elmotasem, Associate Professor, Pharmaceutical Technology Department, Pharmaceutical and Drug Industries Research Institute, National Research Centre, , Egypt . Ph.D, master's degree, and Bachelor's degree from the Faculty of Pharmacy, Cairo University, Egypt. General Certificate of Education. University of London, School Examination Board. Has contributed to 19 scientific papers published in international journals (indexed in Scopus/Web of Science) (H-index 14) and authored two book chapters for Taylor & Francis/CRC Press and Elsevier on functional chemicals and advanced pharmaceuticals. Applied for three patents in the field of diverse drug delivery systems. Additionally, acted as a peer reviewer for leading international publishers including Elsevier, Springer Nature, Bentham, and Dove Press having reviewed over 30 manuscripts. Also co-supervised PhD theses and participated in more than eleven research projects focused on the development of pharmaceutical dosage forms and drug delivery systems. Shared in teaching training courses on cosmetics and drug delivery systems.

Nanoparticles decorated with targeting ligands for precision medicine.

The absence of effective treatments for chronic diseases can lead to treatment failures. Functionalizing nano-delivery systems with targeting ligands presents a potentially advantageous strategy for improving the diagnosis and treatment of chronic diseases. Nanocarriers, encompassing liposomes and other lipidic and polymeric nanocarriers, can be surface-decorated with targeting ligands. Several ligands, including folic acid, peptides, and antibodies, can be used to precisely recognize receptors overexpressed on the surface of diseased organ cells. This study aims to improve the therapeutic effectiveness of natural therapies with reported protective effects but that suffer from limited bioavailability. Our approach focused on developing surface-decorated nano-drug delivery systems specifically targeting certain chronic diseases. Nano-drug delivery systems were developed. They were evaluated for their physicochemical parameters, including drug entrapment efficiency, particle size, surface charge, and drug release. In addition, biochemical markers and histopathological evaluations were conducted in an animal model of certain chronic ailments. Administration of the functionalized targeting formulation resulted in therapeutic improvements relative to the conventional untargeted nano-carrier counterparts. The therapeutic enhancement can be attributed to the targeting characteristics of functionalizing ligands that are directed to inflammatory receptors overexpressed in certain chronic diseases. This approach facilitates the selective delivery of therapeutic agents to certain organs, overcoming biological barriers. As a result, it increases the localized concentration of the therapeutic agent while reducing systemic side effects. This highlights the potential of this strategy to improve the prognosis of different chronic diseases. This approach holds significant promise for advancing precision medicine

Dr. Ahmed Hegazy is a dedicated Researcher at the National Research Centre, specializing in the development and characterization of advanced building materials. With a Ph.D. in Analytical and Inorganic Chemistry, his multidisciplinary expertise bridges the gap between fundamental chemical analysis and practical civil engineering applications.

His core research focuses on sustainable construction materials, non-traditional cements, structural ceramics, and advanced composites. A central theme of Dr. Hegazy's work is the transition toward green engineering, specifically through the formulation of innovative geopolymers and eco-friendly cement alternatives.

Over a career spanning more than a decade, Dr. Hegazy has made significant contributions to the valorization of industrial and natural byproducts. His research extensively covers recycling granite sawing sludge, iron slag, and ornamental stone waste into high-performance bricks, sustainable ceramic tiles, and self-leveling mortars. Furthermore, he has pioneered studies on dual-functional polymer-modified magnesium oxychloride cements and supersulphated cement pastes, optimizing their physico-mechanical properties and durability under adverse environmental conditions. His foundational work also deeply explored the modification of gypsum plaster composites using silica and waste additives to enhance water resistance and compressive strength.

Dr. Hegazy's scholarly impact is reflected in his extensive publication record in premier peer-reviewed journals, including the Journal of Building Engineering, Construction and Building Materials, and Scientific Reports. Through his continuous exploration of hybrid geopolymeric composites and pozzolanic materials, Dr. Hegazy remains at the forefront of developing resilient, sustainable construction solutions that address both environmental challenges and the rigorous demands of modern infrastructure.

Beneficiation of Granite Waste in Sustainable Construction Materials

Granite extraction and processing generate large quantities of fragments, powder, and sawing sludge that are commonly landfilled, creating environmental and resource-management challenges. Owing to their quartz, feldspar, mica, and iron-bearing mineral contents, these wastes can be beneficiated as secondary raw materials in ceramic and cementitious products. In fired clay bricks, granite sludge acts as a non-plastic filler and high-temperature flux. At about 15% replacement, it reduces drying shrinkage and, when fired at 1050 °C, markedly improves strength through liquid-phase formation, particle bonding, and densification. At lower firing temperatures, however, incomplete vitrification may limit mechanical performance.

Granite waste is also suitable for ceramic tiles. A body containing 40 wt% granite waste and kaolin, fired at 1200 °C, developed low water absorption, low porosity, high flexural strength, and good chemical durability. Excessive waste content or firing temperature may cause deformation and bloating because of excessive melt formation and gas entrapment. Aplitic granite waste can additionally replace feldspar and quartz, while iron-rich fractions can be used in self-glazed ceramics and temmoku-type glazes.

In cementitious systems, ornamental-stone sludge can replace part of the natural fine aggregate in polymer-modified self-leveling mortars, improving packing, hydration, and compressive strength. Comparable igneous-rock powders may also partially replace cement and enhance durability.

Overall, granite waste can function as filler, flux, aggregate substitute, glaze precursor, and supplementary cementitious material. Effective utilization depends on matching mineralogy, particle size, replacement level, and processing conditions to the intended product, thereby reducing landfill disposal and conserving natural resources, while supporting circular construction-material manufacturing practices.

Dr. Rasha Mahmoud Abd El- Wahab, Associate Prof. of applied physical chemistry, physical chemistry department, Advanced Materials Technology and Mineral Resources Research Institute, National Research Centre. My research is deeply engaged in materials science and nanotechnology, with a strong focus on environmental sustainability and green applications. Her scientific interests span Recycling environmental waste to produce smart materials and biofuels, the purification of iron-rich ores, the design of advanced nanocomposites for wastewater treatment, and the development of eco-friendly catalysts for biodiesel production. By designing multifunctional materials they contribute to solving pressing challenges in water remediation, renewable energy, and pollution control. Their work reflects a commitment to harnessing innovative nanomaterials not only for energy production and biomedical uses but also for protecting the environment and promoting sustainable technologies.

Evaluation of a sustainable zeolite/bacterial cellulose nanocomposite membrane as a catalyst for biodiesel production from waste cooking oil.

A bacterial cellulose zeolite nanocomposite membrane was successfully prepared, characterized, and evaluated as a heterogeneous catalyst for the transesterification of waste cooking oil into biodiesel. Bacterial cellulose was produced via microbial isolation of Komagataeibacter species, while the zeolite component was synthesized from locally sourced Egyptian raw materials. The prepared catalyst demonstrated efficient catalytic activity, achieving a conversion yield of 77% under optimized reaction conditions: 10 wt% catalyst relative to oil, a methanol-to-oil molar ratio of 40:1, and a reaction time of 9 hours. The physicochemical properties and morphology of the synthesized catalyst were characterized using X-ray diffraction (XRD), scanning electron microscopy (SEM), energy-dispersive X-ray spectroscopy (EDX), and zeta potential analysis. The successful formation of biodiesel was confirmed using thin-layer chromatography (TLC) and high-performance thin-layer chromatography (HPTLC), which were also employed for qualitative and quantitative analysis, respectively. The composition of fatty acid methyl esters (FAMEs) was identified using gas chromatographyâ€“mass spectrometry (GCâ€“MS), revealing the presence of methyl palmitate (40.2%), methyl oleate (53%), and methyl linoleate (5.4%). The produced biodiesel exhibited several fuel properties consistent with international standards, including a saponification value of 193.3 mg KOH/g, an iodine value of 54.6 g Iâ‚‚/100 g, a cetane number of 62.26, and a higher heating value of

42.3 MJ/kg. However, certain properties, such as viscosity and acidity, still require further optimization, despite showing noticeable improvement compared to the original feedstock oil.

Session 5 Materials-Engineering:

Wednesday, 9^{th} September: 3.00 pm-5.00 pm (2h), National Research Center.

Dr. Safaa Hussein Alagamawy (S. H. M. Alagamawy), chemical researcher at the Radio Engineering Research Center. PhD student in the restoration of stone archaeological monuments. Master's degree in the same field, with a thesis entitled "[A Study of Some Multifunctional Nanomaterials in the Restoration and Conservation of Limestone Monuments With Application on a Selected Monument, and a grade of [A+] for [2023].

Evaluating the efficacy of some barium nanocomposites in treating biomicrite limestone .

Biomicrite is one of the most common types of limestone used in archaeological architecture, and it is organic carbonate sedimentary rock, as microfossils and metabolic residues of extinct animals are filled with fine-grained calcite and/or dolomite are scattered in the very fine-grained matrix (micrite), which makes it a relatively weak sedimentary stone, and its preservation is a difficult challenge facing conservators, so they resort to using compatible consolidate materials with limestone components. Nanocomposites are a recent trend in the treatment and conservation of monuments, as they aim to perform a number of functions at the same time, with minimal external chemicals interference. This experimental study aims to evaluate the efficiency of a number of multifunctional nanocomposites, namely nano zinc titanate added to nano barium hydroxide or nano barium titanate or nano barium oxide mixed in paraloid B-72 dissolved in high purity acetone, by examining the consolidate stones samples using stero microscope, PM, SEM, XRD and water contact angle, in addition to measuring the mechanical properties and colorimetric measurement, it was found that nano zinc titrate added to nano barium oxide mixed with paraloid B-72 dissolved in high pure acetone gave the best results.

Dr. Ahmed Mahdy Hamed Haggar is a researcher at the Egyptian Petroleum Research Institute (EPRI), Egypt, in the Process Development Department. He received his Ph.D. in 2022 for his research on the co-production of hydrogen and carbon nanotubes from polyethylene plastic waste. His research interests focus on the catalytic conversion and upcycling of plastic waste into high-value carbon nanomaterials, including carbon nanotubes, carbon nanofibers, graphene and related nanostructures. His work also covers heterogeneous catalysis, hydrogen production, advanced carbon materials, environmental remediation, energy-storage materials, and antibacterial nanomaterials. Dr. Haggar has extensive experience in the development of catalytic processes for waste-to-value conversion and has investigated the transformation of polyolefin wastes into functional carbon nanomaterials with controlled structures and properties. His recent research emphasizes the design of carbon nanostructures with mechanically active surfaces for antimicrobial applications, contributing to sustainable approaches for simultaneous plastic-waste management and advanced-materials production. He has contributed to scientific publications, research projects, and international conferences in the fields of nanomaterials, catalysis, hydrogen production, waste valorization, and environmental applications.

Catalytic Upcycling of Plastic Waste into Carbon Nanoblades and Carbon Nanodarts with Enhanced Mechano-Bactericidal Activity

The increasing accumulation of plastic waste and the emergence of antibiotic-resistant bacteria have created an urgent need for sustainable strategies that simultaneously address environmental and healthcare challenges. This study presents a waste-to-value approach for converting plastic waste into high-value carbon nanostructures with potent antibacterial activity. Two distinct carbon nanomaterials, namely carbon nanoblades and carbon nanodarts, were synthesized through catalytic carbonization under controlled thermal and atmospheric conditions. The resulting nanostructures were systematically characterized to establish their morphology, crystallinity, and graphitic nature, and their antibacterial performance was evaluated against representative Gram-positive and Gram-negative bacterial strains. Transmission electron microscopy revealed the successful formation of platelet-like carbon nanofibers with sharp blade-like edges (carbon nanoblades) and elongated tubular carbon nanotubes (carbon nanodarts). X-ray diffraction and Raman spectroscopy confirmed the graphitic structure of both materials, while carbon nanoblades exhibited a slightly higher defect density than carbon nanodarts, suggesting the presence of more active surface sites. Antibacterial activity was assessed against Staphylococcus aureus, Escherichia coli, and Pseudomonas aeruginosa using colony-count and inhibition-zone assays. Both carbon nanostructures demonstrated significant bactericidal effects toward all tested strains. However, carbon nanoblades consistently exhibited superior antibacterial performance compared with carbon nanodarts. The enhanced activity of carbon nanoblades is attributed to their sharp-edged morphology and higher surface roughness, which facilitate direct physical damage to bacterial membranes through a mechano-bactericidal mechanism, leading to membrane disruption and cell death. These findings demonstrate an effective route for upcycling plastic waste into functional carbon nanomaterials and highlight carbon nanoblades as promising antibacterial agents for environmental, biomedical, and antimicrobial surface applications.

Asmaa O. Ahmed is an Assistant Researcher in the Physical Chemistry Department at the National Research Centre and a PhD candidate whose work sits at the intersection of materials science and clean energy technology. Her research centers on electrochemical energy applications, with particular emphasis on water electrolysis for green hydrogen production and the development of high-performance supercapacitors central to the shift toward sustainable energy storage and conversion.

Her approach combines hands-on expertise in material synthesis, using techniques including sol-gel, hydrothermal, co-precipitation, and electrochemical preparation methods, with a strong foundation in structural and electrochemical characterization.

Asmaa's doctoral research investigates perovskite oxide materials, particularly $SrFeO_3$ and $SrNiO_3$, as electrocatalysts for water electrolysis, aiming to improve the efficiency and stability of catalysts critical to scalable green hydrogen production.

Role of Oxygen Vacancies in Enhancing the Hydrogen Evolution Reaction Performance of $SrFeO_{3-x}$ Perovskite Oxides

Perovskite oxides have emerged as promising electrocatalysts for the hydrogen evolution reaction (HER) due to their structural flexibility, tunable electronic properties, and defect engineering capability. In this work, we investigate the influence of spin states and oxygen vacancies on the HER activity of the transition-metal perovskite SrFeO3-X. We successfully synthesized four SrFeO3-X samples by facile sol gel technique through changing either calcination time. XRD revealed that the two samples preheated for 2 h at 500 oC then calcined for 5 and 10 h at 900 oC have a cubic phase mixed with a small percentage of orthorhombic phase. While the other two samples have pure cubic phase. SEM indicated the porous structure of both samples. In contrast to XRD, Electrical properties, Roman spectroscopy, and ESR exposed that all samples contain mixed phases with different percentage. Where, ESR analysis (g of all samples ≈ 2.00) indicates defect-related paramagnetic centers in SrFeO3-X, with optimized oxygen vacancy concentration (S2) correlating with enhanced hydrogen evolution reaction performance. S2 showed the lowest overpotential of 420 mV vs RHE, with best stability over 41 h. Perovskite oxides with mixed-valence Fe3+/Fe4+ states, strong Fe–O–Fe interactions and controlling oxygen vacancy, contribute to coupled electronic and local magnetic behavior, hence, strongly affect electrocatalytic catalytic performance.

Dr. Adel A. Koriem is an Egyptian researcher and Assistant Professor in the Department of Polymers and Pigments at the National Research Centre (NRC), Egypt. Born in Giza, Egypt, in 1972, he has built an extensive scientific career spanning more than two decades in polymer science, rubber technology,materials characterization, and nanotechnology.

Dr. Koriem began his professional career at the National Research Centre in 1999 as an Assistant Researcher in the Department of Polymers and Pigments. In 2004, he became a Research Assistant and continued developing his expertise in polymer and rubber materials. Since 2010, he has worked as a Researcher in the Department of Polymers and Pigments within the Division of Chemical Industries, and since June 2025 he has served as an Assistant Professor.

His research activities focus particularly on polymer engineering technology and nanotechnology, including the synthesis and characterization of advanced polymeric and rubber materials. His work involves investigating the chemical structures and properties of synthesized materials, developing antimicrobial and fire-retardant rubber and polymer products, and exploring sustainable materials and nanocomposites. He has also contributed to the development of biochar- and agricultural-waste-derived materials as reinforcing fillers for rubber composites.

Dr. Koriem has an extensive record of scientific publications covering topics such as NBR and EPDM rubber vulcanizates, antimicrobial rubber materials, fire-retardant polymeric materials, sustainable fillers, biochar-based composites, natural antioxidants, cellulose nanocrystals, and advanced rubber nanocomposites. His recent publications include studies published in journals such as *Polymer Composites*, *Radiochimica Acta*, *BMC Chemistry*, *and Materials Science: Materials in Electronics.*

In addition to his research publications, Dr. Koriem is an inventor on a patent entitled "Safety and Antimicrobial Rubber Product and the Method of Preparation," registered under Certificate No. 31131 in 2023.

Dr. Koriem has actively participated in international scientific conferences, including MACRO IUPAC in Istanbul, the European Polymer Congress in Dresden, the Arab International Conference on Polymer Science and Technology in Italy, and an international conference on nanostructured polymers and nanocomposites in Dresden.

He has also contributed to several research projects funded by the National Research Centre, covering polymer-clay nanocomposites, fire-retardant polymeric materials, antimicrobial rubber products for children's safety toys and medical applications, and green antimicrobial rubber products for personal protection.

Beyond academia, Dr. Koriem has served as a scientific consultant for numerous industrial organizations in Egypt, including companies working in plastics, rubber, automotive, chemical, medical-supply, packaging,and manufacturing sectors. He is also a reviewer for academic journals, including Polymer *Bulletin* and The *Egyptian Journal of Chemistry*, and is a member of several professional scientific societies related to chemistry, polymer science, advanced materials, and nanotechnology

His technical expertise includes Differential Scanning Calorimetry (DSC), Fourier Transform Infrared Spectroscopy (FTIR), tensile testing, Particle Charge Detection (PCD), water and oxygen vapor permeability measurements, and Gel Permeation Chromatography (GPC). He is proficient in Microsoft Office and has experience with several scientific and computing programs.

Overall, Dr. Adel A. Koriem's career reflects a strong combination of academic research, applied materials science, industrial consultancy, scientific publication, and innovation. His work contributes particularly to the development of safer, antimicrobial, fire-retardant, sustainable, and high-performance polymer and rubber materials.

Interfacial Engineering of CuAlZn Alloy-Reinforced NBR/EPDM Elastomers for Soft Magnetic, Dielectric, and Thermomechanical Applications

This study reports the fabrication of multifunctional NBR/EPDM elastomer composites incorporating a CuAlZn alloy as a functional filler together with 3-(trimethoxysilyl)propyl methacrylate (TMSPM) to improve interfacial compatibility. The designed materials were intended to introduce soft magnetic functionality while preserving favorable curing characteristics and mechanical, thermal, and dielectric performance. Alloy contents ranging from 2.5 to 35 phr were investigated through rheological measurements, stress relaxation analysis, differential scanning calorimetry (DSC), thermogravimetric analysis (TGA), tensile testing, scanning electron microscopy (SEM), vibrating sample magnetometry, and broadband dielectric spectroscopy. The silane coupling agent strengthened the polymer filler interface, resulting in homogeneous particle distribution at intermediate filler concentrations. Composites containing moderate alloy loadings, particularly NE2 and NE3, exhibited superior vulcanization behavior, enhanced tensile properties, lower damping behavior (tan Î´), and reduced solvent uptake, indicating the formation of a more tightly crosslinked and mechanically robust network. Thermal characterization demonstrated improved molecular organization and greater resistance to thermal degradation following alloy incorporation. Magnetic measurements confirmed the successful development of soft magnetic elastomers, with NE3 delivering the most advantageous combination of magnetic response and structural performance. Incorporation of the metallic alloy also enhanced dielectric permittivity and electrical conductivity, where NE3 showed a pronounced temperature-dependent dielectric response, whereas NE8 achieved the greatest conductivity and the highest predicted electromagnetic interference (EMI) shielding capability at elevated frequencies. Nevertheless, excessive alloy loading promoted particle agglomeration, leading to deterioration of mechanical homogeneity. Collectively, these findings demonstrate that the synergistic combination of CuAlZn alloy and TMSPM enables the production of multifunctional elastomeric composites possessing improved mechanical strength, thermal stability, magnetic responsiveness, and dielectric performance, making them promising candidates for flexible magnetic devices, adaptive vibration-control systems, and lightweight EMI shielding materials.

Dr. Rasha Mostafa Al-Makhlasawy is a researcher at the Digital Signal Processing Laboratory, Computers and Systems Department, Electronics Research Institute (ERI), Cairo, Egypt. She received her Ph.D. in Electronics and Electrical Communications Engineering from the Faculty of Engineering, Tanta University, Egypt, in 2020.

Her research focuses on artificial intelligence, machine learning, deep learning, digital signal processing, wireless communications, biomedical signal and image processing, Internet of Things (IoT), and intelligent healthcare systems. She has contributed to numerous interdisciplinary research projects that integrate advanced AI techniques with signal processing to address real-world challenges in healthcare, smart communications, and intelligent sensing.

Dr. Al-Makhlasawy has authored and co-authored several peer-reviewed scientific publications in international journals and conferences. Her current research interests include explainable artificial intelligence (XAI), reinforcement learning, computer vision, medical image analysis, digital twins, and next-generation communication systems, particularly 6G-enabled intelligent networks. She is also actively involved in developing AI-driven solutions for disease diagnosis, predictive analytics, and smart decision-support systems.

In addition to her research activities, she is passionate about promoting STEM education and empowering young researchers through scientific training programs, workshops, and collaborative research initiatives. She actively supports interdisciplinary collaboration and encourages the participation of women in science, technology, engineering, and mathematics.

Artificial Intelligence for Next-Generation Wireless Communications: Enabling Intelligent, Adaptive, and Sustainable Networks

Artificial Intelligence (AI) is transforming wireless communication systems by enabling networks to become more intelligent, adaptive, and autonomous. As wireless technologies evolve toward sixth-generation (6G) networks, traditional communication techniques face

significant challenges in meeting the increasing demands for ultra-high data rates, ultra-low latency, massive device connectivity, and energy-efficient operation. AI offers innovative solutions by integrating machine learning, deep learning, and reinforcement learning into communication system design and network management.
This presentation highlights the growing role of AI in enhancing the performance of modern wireless communication systems. It discusses how AI-driven algorithms improve channel estimation, resource allocation, spectrum management, beamforming, interference mitigation, modulation adaptation, and network optimization. Special attention is given to reinforcement learning techniques that enable communication systems to learn from dynamic environments and make intelligent real-time decisions without relying solely on predefined mathematical models.
The presentation also explores the integration of AI with emerging technologies, including the Internet of Things (IoT), edge computing, digital twins, reconfigurable intelligent surfaces (RIS), and intelligent transportation systems. These technologies collectively contribute to building self-organizing and self-healing wireless networks capable of supporting future smart cities, healthcare, industrial automation, and autonomous vehicles.
Despite its significant potential, several challenges remain, including data privacy, computational complexity, model interpretability, cybersecurity, and energy consumption. Addressing these challenges requires interdisciplinary collaboration between communication engineers, AI researchers, and industry stakeholders.
Overall, AI is expected to become a fundamental component of future wireless communication infrastructures, driving the evolution toward intelligent, resilient, and sustainable networks while creating new opportunities for scientific research, technological innovation, and real-world applications across diverse sectors.

Dr. Ahmed Ghanem is an assistant professor at the Packaging Materials Department, National Research Centre (NRC), Egypt, and a former Postdoctoral Researcher at the Institute of Hydrogen Energy, Korea Institute of Energy Technology (KENTECH), South Korea. He received his Ph.D. in Polymer Chemistry from Ain Shams University in 2021. Throughout his academic career, he conducted research visits and internships at the University of Arkansas (USA), Université Claude Bernard Lyon (France), the Institute of Physical Chemistry of the Polish Academy of Sciences (Poland), and the University of Science and Technology of China (USTC). Dr. Ghanem has authored more than 33 peer-reviewed publications, is the inventor of two patent applications, H-index 19, and has participated in numerous national and international research projects as a PI and member. He serves as a reviewer for leading scientific journals and has received several awards for outstanding research and scientific presentations. His research interests focus on polymeric materials and nanocomposites for sustainable applications, particularly in food packaging, water treatment, energy storage and conversion, and biomedical technologies.

Development of PVA Composite Films for Meat Preservation

Meat products are the most common sources of protein. However, meat products are highly prone to spoilage during storage or even transportation. Several reports are released annually, presenting promising solutions to overcome meat preservation challenges. In this work, novel composite polyvinyl alcohol (PVA) films have been developed and investigated for packaging fresh meat. Specifically, active components were extracted from hibiscus flowers and encapsulated using a spray dryer. The well-assembled microcapsules were incorporated into PVA matrices at different concentrations to obtain free-standing composite films. FTIR, SEM, Zetasizer, TGA, mechanical testing, water contact angle measurements, and permeability were performed. Significant changes in the physicochemical and biological properties of PVA films were observed, thanks to the successful formation of extract microcapsules and their inclusion in the polymer matrix. Moreover, the results emphasized the superiority of composite PVA films containing microcapsules over free-form films as packaging materials for real meat samples. The results showed substantial antioxidant activity, along with prolonged antibacterial activity, evidenced by a decline in bacterial growth and a near-pH plateau during refrigerated storage (4 °C) for up to 10 days. These findings support the sustained-release mechanism, which extends their biocidal effect over the storage time. This highlights that adding stable microencapsulated extracts into biopolymers can enhance the applicability and workability of natural extracts for use as active packaging materials, overcoming the limited performance of traditional films incorporating free extracts.

Dr. Basant Aboelsaud Elsebai is a researcher at the National Research Centre (NRC), Egypt, with expertise in analytical chemistry, electrochemistry, nanomaterials, sensors, and wastewater treatment. She obtained her B.Sc. in Chemistry from Cairo University in 2011, followed by an M.Sc. in Analytical Chemistry in 2020. During her early research career, she contributed to a Marie Curie joint project on electroanalytical sensors for heavy metals and biological species and undertook an international research internship at the University of Coimbra, Portugal, under Prof. Christopher Brett. She received the NRC Best International

Publication Award for Young Researchers in 2019 and completed her PhD in Wastewater Treatment Applications at Ain Shams University in 2024. Dr. Elsebai has participated in several national and international research projects and conferences and has published multiple peer-reviewed international articles. Her current research focuses on nanomaterial-based sensors and biosensors, electrochemical technologies, and the development of low-cost, high-efficiency materials for innovative wastewater treatment applications.

Zeolite-Infused Nanofibrous Hydrogel for Water Treatment Applications

Heavy metals pose considerable challenges for the global community. Several studies are released annually, presenting new techniques or materials to mitigate this environmental issue. However, many shortcomings are still associated with this kind of research in terms of performance, durability, and cost. Therefore, in this study, a nanofibrous polysaccharide-based hydrogel was produced in a fermentation medium. In an innovative approach, the isolated bio-mat was modified with Na-zeolite particles derived from Egyptian kaolin, resulting in a fully renewable bionanocomposite. This unique assembly is theoretically attractive and exhibits significantly different properties from those obtained by conventional methods. The modified hydrogel was extensively characterized, and the results showed successful anchoring of crystallized zeolite particles within the hydrogel's open cells, displaying improvements in surface texture and diverse surface charges. Due to its superior properties, the bionanocomposite was tested as an adsorbent for metal ions in synthetic wastewater. Within minutes, the bionanocomposite effectively demineralized the wastewater, achieving high adsorption capacities of 77.33, 76.39, and 24.06 mg g^{-1} for Pb (II), Cd (II), and Ni (II) ions, respectively. Kinetic and isotherm models indicated that the adsorption followed a pseudo-second-order reaction and obeyed Langmuir isotherms. Dubinin's analysis demonstrated that the removal mechanism is chemisorption. Thermodynamic studies confirmed the exothermic nature of the reaction. Durability tests showed that the modified hydrogel could be reused over five cycles. This research offers an opportunity to develop new adsorbents with both economic and environmental benefits for sustainable water treatment technologies.

Dr. Ahmed Nabile Emam Osman, earned his BSc and MSc degrees in medical biophysics in 2006 from Helwan University, and 2010 from Cairo University, respectively. In October 2006, he joined the Center of Excellence for Advanced Sciences - National Research Center (NRC), where he was engaged in Advanced Materials & Nanotechnology Lab as a temporary contract till his graduation of MSc Degree. In November 2017, he was awarded the PhD degree in Laser applications in Photochemistry and Photobiology from the National Institute of Laser Enhanced Sciences – Cairo University. His research work in MSc and PhD focused on both of the biophysical study of semiconductor nanoparticles doped with different magnetic materials to be used in biomedical applications, and development of Fluorescent hybrid nanocomposites for biomedical imaging and early diagnosis of cancer, under supervision of Prof. Amani Mostafa the professor of materials science and Former Dean of Advanced Materials Technology & Mineral Resources Research Institute – National Research Centre, and Prof. Mona Bakr Mohamed, who she was the former postgraduate student of Prof. Mostafa A. El-Sayed, and the Horney professor of nanotechnology and ultrafast spectroscopy and the executive director the Egyptian Nanotechnology Center (EGNC) – Cairo University. In Sep. 2012, he joined to Ultrasound Imaging and Therapeutics Research Laboratory, Biomedical Engineering Department, University of Texas at Austin – USA as a Visiting Researcher till Jan. 2013. In this period, his research work focused on engineering nanomaterials to be used as Photo-acoustic Tomography Imaging (PAT). Dr. Emam published about **46** archival publications (**42** Peer reviewed articles and **4** Book Chapters). Also, he is a reviewer in 30 Scopus indexed journals The main research of interest is based on the material science and nanotechnology, where this research was focused on the fabrications of different categories of nanomaterials such as: Magnetic, Plasmonic, Semiconductor, Metal Oxide, Hybrid Nanocomposites (i.e.; Nanoalloys, Doped Nanomaterials), Carbon –based nanomaterials such as, Carbon quantum dots (C-Dots) and their hybrid nanocomposites. In addition, investigation of the photophysical properties of nanomaterials and their hybrid nanostructures such as, magneto-plasmonic nanoalloys, doped semiconductors, carbon-based nanomaterials. Furthermore, his research focused on the use of nanomaterials in the water treatment, archaeological, antimicrobial, anti-fouling applications. Moreover, Green synthesis of nanomaterials, Energy applications of nanomaterials. Ahmed is also deeply engaged in academic capacity-building through workshops and international programs on nanotechnology, quantum dots, and research ethics across Egypt, Africa, and beyond. His training includes degrees in Medical Biophysics and Laser Applications in Photochemistry and Photobiology, complemented by advanced courses in AI for scientific research, peer review, and statistics. Alongside his research, he mentors graduate students, develops industry-oriented nano-enabled solutions, and contributes book chapters on toxicity and environmental sustainability of hybrid plasmonic nanocomposites.

Engineered Hybrid Nanocomposites for Safer Healthcare, Cleaner Water, and Sustainable Heritage Preservation

Engineered hybrid nanocomposites are emerging as multifunctional platforms that connect nanomedicine, environmental remediation, and cultural heritage conservation to enable a safer and healthier life-style. By integrating plasmonic, magnetic, semiconductor, metal-oxide, carbon-based, and biopolymer building blocks, these nanoarchitectures are tailored for cancer theranostics, antimicrobial control, bone regeneration, and advanced imaging and sensing. Representative systems include gold-doped wollastonite for osteogenic repair, multifunctional silver, copper oxide, and zinc oxide hybrids to suppress multidrug-resistant pathogens, and carbon dot–based nanocomposites for high-contrast diagnostic imaging and latent fingerprint visualization. In parallel, water-desalination and photocatalytic nanomaterials, nanostructured consolidants for limestone and stucco monuments, and nano-enabled inks and coatings for industrial and biomedical interfaces collectively target generous quality-of-life improvements. Emphasis on green synthesis, toxicological evaluation, and application-relevant performance underpins their translation into theranostic healthcare, clean-water technologies, and sustainable conservation strategies. Altogether, engineered hybrid nanocomposites constitute an integrated toolbox for advancing health security, environmental safety, and heritage durability, thereby supporting more generous and healthier lifestyles at individual, societal, and ecosystem levels.